

Solution

FORCE AND PRESSURE -02

Class 08 - Science

Section A

1. Read the text carefully and answer the questions:

Students of class 8 playing tug of war in the playground. Suddenly the teacher came and called all the students and started telling some facts related to the game. That the two teams pull the rope in opposite directions. Members of both the teams try to pull the rope in their direction. Sometimes the rope simply does not move.



- (i) (c) difference between two forces

Explanation: difference between two forces

- (ii) (a) difference between the two forces

Explanation: difference between the two forces

- (iii) (a) Push or pull

Explanation: Push or pull

- (iv) (b) False

Explanation: False

- (v) 1. magnitude

2. Read the text carefully and answer the questions:

Teacher of class 8 performed the following activity in the class.

Step 1 - Take a plastic bottle. Fix a cylindrical glass tube, a few cm long near its bottom

Step 2 - Cover the mouth of the glass with a thin rubber sheet.

Step 3 - Now fill the bottle up to half with water.

Step 4 - Some changes occur. Pour some more water in the bottle.



- (i) (b) Water exerts pressure

Explanation: Water exerts pressure

- (ii) (b) Increases with depth

Explanation: Increases with depth

- (iii) (b) Container exerts pressure on the side of wall

Explanation: Container exerts pressure on the side of wall

- (iv) 1. pressure

- (v) (a) True

Explanation: True

3. Read the text carefully and answer the questions:

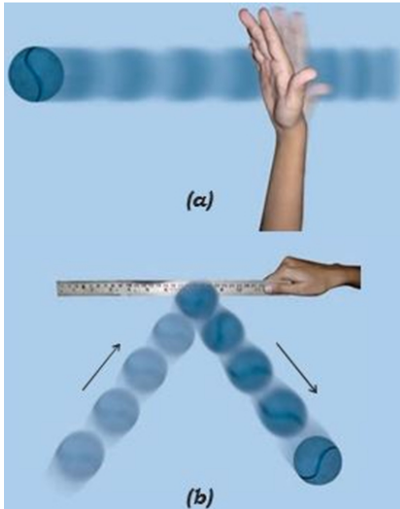
Step 1 - Take a ball and place it on a level surface.

Step 2 - Make the ball move by giving it a push.

Step 3 - Now place the ruler from your geometry box in its path.

Step 4 - Does the ball continue to move in the same direction after it strikes the ruler

Step 5 - In each case note your observations about the direction of motion of the ball after it strikes the ruler.



- (i) **(d)** Zero

Explanation: Zero

- (ii) **(b)** all of these

Explanation: all of these

- (iii) **(a)** Both position of rest and position of motion

Explanation: Both position of rest and position of motion

- (iv) 1. increases

- (v) **(a)** True

Explanation: True

4. Read the text carefully and answer the questions:

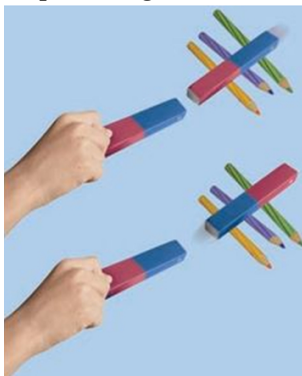
There are 3 type of non-contact forces following activity describe one of them.

Step 1 - Take a pair of bar magnets. Place the longer side of one of the magnets over three round shaped.

Step 2 - Bring one end of the other magnet near the end of the magnet placed on the rollers.

Step 3 - Make sure that the two magnets do not touch each other.

Step 4 - Bring the other end of the magnet near the same end of the magnet placed on the rollers.



- (i) **(c)** Magnetic force

Explanation: Magnetic force

- (ii) **(b)** electrostatic force

Explanation: electrostatic force

- (iii) **(a)** force of gravity

Explanation: force of gravity

- (iv) 1. non-contact force

- (v) (a) True

Explanation: True

5. Read the text carefully and answer the questions:

Ankit is studying about contact forces. Firstly, he study about muscular force that it is the muscular force that enables us to perform all activities involving movement or bending of our body. All our body process is possible by muscular force. Muscular force of animal is used to carry out work. Secondly, he study about frictional force.



- (i) (d) Opposite

Explanation: Opposite

- (ii) (a) All of these

Explanation: All of these

- (iii) (d) muscular force

Explanation: muscular force

- (iv) 1. muscular

- (v) (a) True

Explanation: True

Section B

6.

- (c) contact force

Explanation: contact force

7.

- (b) All of these

Explanation: All of these

8.

- (c) 490 N

Explanation: Here, Pressure exerted by the boy on ground = 7N/cm sq

Area on the boy occupied = 70 cm sq

As, Pressure = Force/ Area

The force acting on the ground = Pressure $\times$ Area

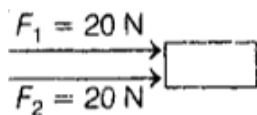
$$= 7 \text{ N/ cm sq} \times 70 \text{ cm sq}$$

$$= 490 \text{ N}$$

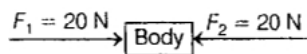
9. (a) 40 N and 0

Explanation: When two forces act in the same direction and parallel to each other; then the net force is

$$F_{\text{net}} = 20\text{N} + 20\text{N} = 40\text{N}$$



When two forces act in the opposite direction as shown in the figure.



$$F_{\text{net}} = F_1 + F_2$$

Here, $F_1 = 20\text{ N}$

$F_2 = -20\text{N}$ (opposite direction)

$$= 20 - 20 = 0$$

10. **(b)** 1 kgf = 9.8 N
Explanation: 1 kgf = 9.8 N
11. **(b)** 746 W
Explanation: The electrical equivalent of one horsepower is 746 watts in the International System of Units (SI), and the heat equivalent is 2,545 BTU (British Thermal Units) per hour. Another unit of power is the metric horsepower, which equals 4,500 kilogram-metres per minute (32,549 foot-pounds per minute) or 0.9863 horsepower.
12. **(a)** Newton
Explanation: S.I. unit of force is Newton. When a force pushes a body of mass one kg to a distance of one meter per second, the force is equal to one Newton.
13. **(b)** Dyne
Explanation: Dyne
14. **(d)** Direction as well as speed
Explanation: When a batsman hits the ball, it changes the direction of ball as well as speed of the ball. Due to the force applied by the bat along the direction of motion the speed of the ball increases.
15. **(d)** 100 Pa
Explanation: Here, Force = 200N, Area = 2 sq m.
 Pressure = Force/Area.
 Pressure = 200N / 2 sq m = 100 Pascal (Pa).
16. **(c)** Overcome frictional force
Explanation: Frictional force always works in opposite direction of motion. A body moving on leveled grounds need to apply constant force to overcome frictional force provided by ground.
17. **(c)** between all bodies in the universe
Explanation: between all bodies in the universe
18. **(a)** Zero
Explanation: The value 'g' at the centre of earth becomes zero.
19. **(d)** N-m^{-1}
Explanation: The SI unit of pressure is pascal. The pressure is given by

$$p = \frac{\text{Force}}{\text{Area}} = \frac{\text{N}}{\text{m}^2}$$
 In C.G.S system the unit of pressure is dyne/cm^2 .
 Thus, N-m^{-1} is not a unit of pressure.
20. **(d)** Drag
Explanation: The frictional force exerted by fluids is also called drag. Water and other liquids exert force of friction when objects move through them. The friction force exerted by fluids depends on its speed with respect to fluid and also on the shape of the object and the nature of the fluid.
21. **(b)** Provide better grips with ground
Explanation: The threaded tyres of truck, cars and buses provide better grips with ground because threads in tyres creates more irregularity to the surface of the road and causes greater frictional force to act and form better grips.
22. **(b)** ground

Explanation: According to Newton's third law; every action has an equal and opposite reaction. This is relevant to walking because when you put your foot on the ground, you are applying a force to it. In doing this, the ground also actually applies an equal force onto your foot, in the opposite direction, pushing you forward.

23.

(b) Friction is absent

Explanation: Frictional force always acts in opposite direction of motion to slow down or stopped the body from moving. A body will remain in state of motion on rolling on ground if friction is absent or the body will not move at all as friction is due air also.

24.

(d) Speed

Explanation: A force applied on an object may change its speed. When force is applied in direction of motion speed will increase and if the force is applied in a direction, opposite to its motion, the speed decreases.

25. **(a)** water reduces the friction between the road and the tyres

Explanation: water reduces the friction between the road and the tyres

26.

(d) 1 kg m/s^2

Explanation: One Newton is equal to 1 kg m/s^2 . One newton is equal to 1 kilogram meter per second squared.

27.

(b) Electrostatic force

Explanation: During dry weather, while combing hair, sometimes we experience hair flying apart. The force responsible for this is electrostatic force. The force exerted by a charged body on another charged or uncharged body is known as electrostatic force.

28.

(b) 1000 kg

Explanation: The weight of air in a column of the height of the atmosphere and area $10\text{cm} \times 10\text{cm}$ is equal to 1000 kg. Atmospheric pressure is the weight exerted by the overhead atmosphere on a unit area of surface.

29. **(a)** Boiling happens at higher temprature

Explanation: Food is cooked more quickly in a pressure cooker because at the higher pressure (1 bar/15 psi), the boiling point of water rises from 100°C (212°F) to 121°C (250°F). The hotter steam is able to transmit its thermal energy o the food at around 4 times the rate of conventional boiling.

30.

(c) option (i)

Explanation: Pressure is a scalar quantity, not a vector quantity. It has a magnitude but no direction associated with it. Pressure acts in all directions at a point inside a gas. At the surface of a gas, the pressure force acts perpendicular to the surface.

31.

(d) 3246.75 Pa

Explanation: Here, Force = 50 N

Radius of circular plate = 7cm

Area of a circle = $\pi r^2 = 3.14 \times 7 \times 7 = 154 \text{ cm}^2 = 154/10000 \text{ m}^2 = 0.0154 \text{ m}^2$.

Pressure = Force/Area = $50\text{N}/0.0154 \text{ m}^2 = 3246.75 \text{ Pa}$

32.

(d) Force of friction

Explanation: When the applied force is in contact with an object, then this is called contact force. Among the given options, the force of friction is a contact force because the force of friction arises due to contact between surfaces.

Magnetic force, the force of gravity, and electrostatic force are non-contact forces.

33.

(c) 2.5kg

Explanation: Here, force = 5 N and acceleration = 2m/s^2 . Force = Mass $\times$ Acceleration, Mass = Force/Acceleration. So, Mass = $5 \text{ N}/2\text{m/s}^2 = 2.5 \text{ Kg}$

34. **(a)** Static friction
Explanation: Static friction is the friction that exists between a stationary object and the surface on which it is resting. Static force should be greater than frictional force in magnitude.
35. **(d)** Height of the column
Explanation: The pressure exerted by water at the bottom of the container depends on the height of the column of water. Pressure of water increases with depth.
36. **(d)** increase friction force acts on it
Explanation: increase friction force acts on it
37. **(b)** 4 kg
Explanation: 4 kg
38. **(d)** the water exerts more pressure on bottom wall
Explanation: the water exerts more pressure on bottom wall
39. **(b)** Thrust
Explanation: The perpendicular force acting per unit area is called Thrust. S.I. unit of thrust is Newton.
40. **(d)** Increased pressure, increases the boiling point
Explanation: The pressure cooker traps that hot air and moisture with the food, which expedites the cooking process. In other words, the moisture surrounding the food itself reaches higher temperatures than it would without the pressure, which speeds up the chemical processes involved in cooking.
41. **(b)** muscular
Explanation: muscular

Section C

42. **(b)** maximum in position A
Explanation: The smaller the area, the larger the pressure on a surface for the same force. Hence, brick A whose area of contact is minimum will exert more pressure on the table.
43. **(b)** B and C
Explanation: In the given water tank, B and C are at the same levels. Hence, from taps B and C, water will flow out at the same pressure.
44. **(a)** Magnetic force alone
Explanation: The magnetic force is responsible for the movement of the hammer. When an electric current flows through the coil and the circuit is completed, the magnetic force is produced. As a result, it acts as an electromagnet.
45. **(b)** Magnitude of force applied by A is greater than that of B.
Explanation: If the block moves towards the right, the magnitude of force applied by A is greater than that of B.
46. **(a)** Pressure at A < Pressure at B < Pressure at C
Explanation: The pressure of water increases with an increase in depth. In the given container, C is in-depth. Hence, its pressure will be more. Therefore, the correct order is:
 The pressure at A < Pressure at B < Pressure at C.

Section D

47. **(c)** Atmospheric pressure
Explanation: Astronauts wear special suits when they go into space because pressure inside the suit is maintained at

atmospheric pressure on earth. As there is less atmospheric pressure in space to maintain the equilibrium astronauts wear special suits.

48.

(b) Equal

Explanation: The pressure inside our bodies is equal to the atmospheric pressure of atmosphere. At high altitudes, the atmospheric pressure decreases but blood pressure remains the same.

49.

(c) Pull

Explanation: To draw water from a well we have to pull the rope in upward direction. The force required to pull the rope depends upon mass of the bucket with water and height in this case it is the depth of the water.

50.

(b) Relative motion

Explanation: Friction opposes the relative motion between two surfaces in contact. It always acts in the direction opposite to the direction of motion of the object. Two moving objects in contact with each other are always in relative motion.

51. **(a) Zero**

Explanation: The force on an object is zero if the two forces acting on it in opposite directions are equal in magnitude. This kind of force is called a balanced force.

52. **(a) Vector**

Explanation: Force is a vector quantity as force contains both magnitude of the medium and direction of the movement the medium has taken. Forces is push or pull acting on body.

53. **(a) Results in a force**

Explanation: An interaction of one object with another object results in a force between the two objects. Force can change the shape, state of motion and direction of the object.

54.

(b) Repels

Explanation: The north pole of magnet repel the north pole of another magnet because like poles of magnet when face each other repel and unlike poles get attracted when each other.

Section E

55.

(b) atmospheric pressure

Explanation: The rise of water in the dropper is due to atmospheric pressure. When the bulb of a dropper is pressed, the pressure inside the dropper decreases and this causes the water to go inside the dropper.